What does a richer colour vocabulary buy us?

An analysis of the World Color Survey

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Same chips. Different names.
What can regression explain?

The questions guiding this project

From counting colour words to understanding how people organise colour space

01 Language-level vocabulary and geometry

Are richer shared inventories associated with finer average distinctions between nearby colours and more compact naming categories?

02 Tail prevalence

Are richer-vocabulary languages associated with different proportions of low- and high-scoring speakers?

03 Demographic heterogeneity and moderation

Do age and recorded-sex associations vary across languages, and does the richness association differ across age and sex?

04 Colour-space localisation

Are richness and demographic associations stronger in particular sampled regions of colour space?

05 Individual-level prediction

Do vocabulary and demographic variables improve held-out prediction of individual partition scores enough to justify added complexity?

From theory to broader evidence



Berlin & Kay

Basic Color Terms (1969) **Question:** Are colour categories entirely language-specific?
Their study of 20 languages proposed 11 basic colour categories and a constrained sequence in which languages tend to acquire additional basic terms.

Contribution: A theory of cross-linguistic regularities in colour naming, the size and representativeness of the original sample motivated wider testing.

World Color Survey

A larger empirical test The WCS extended the investigation to 110 languages and 2,616 informants, using the same standard array of 330 Munsell colour chips.
It recorded individual naming responses and focal-colour selections, allowing comparisons across languages and between speakers of the same language.

Contribution: A much broader dataset for testing recurring patterns while retaining individual variation.

From recurring motifs to the present analysis



Lindsey & Brown

PNAS (2009) Analysing 2,367 WCS informants, they identified recurring colour-naming motifs while finding that several different motifs often occurred within one language.
Their work shifted attention from one label system per language to the diversity of individual speakers' naming patterns.

Implication: Cross-linguistic regularities can coexist with substantial within-language variation.

The present project

Regression Techniques We ask whether vocabulary richness is associated with finer distinctions between nearby chips, more compact categories, and different proportions of low- and high-scoring speakers.
We then examine age, sex, colour region and whether added predictors improve held-out prediction enough to justify their complexity.

Contribution: Quantify associations using course-level regression, diagnostics and model comparison.

What is in the World Color Survey?

110 languages

2,616 informants

330 Munsell chips per speaker

863,280 naming responses in archive

Two main tasks

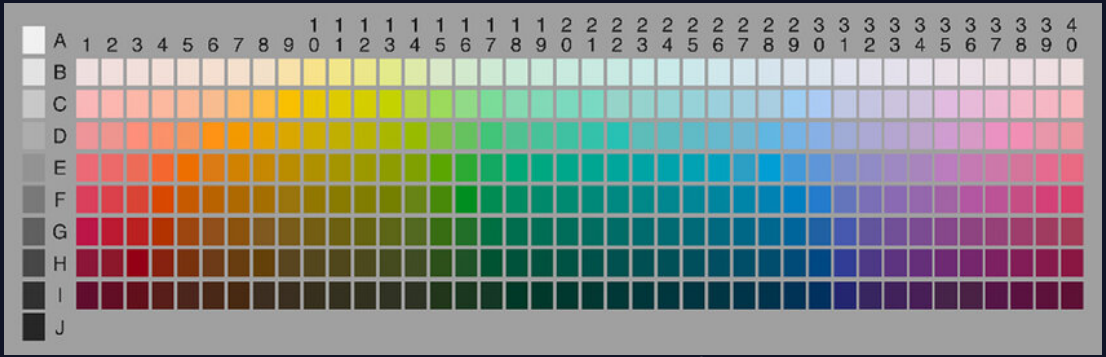
Naming task: For each chip, record the colour term supplied by the speaker.

Focal-colour task: For each major term, select its best example colour chip or chips.

Which speakers enter the analysis?

- **2,447 speakers:** We kept speakers who named at least 300 of the 330 colour chips.
- **109 languages:** After this filter, at least one eligible speaker remained in 109 of the original 110 languages.
- **2,355 speakers:** For the main regression involving age and sex, we further retained only speakers with non-missing outcome, age and sex data.

These are the colour chips used in the study



These are the 330 Munsell colour chips used in the World Color Survey. In the naming task, speakers supplied a colour term for each chip. In the separate focal-colour task, they selected the best example chip or chips for a colour term.

From archival files to analysis variables

Files used	Records × columns
`term.txt` Language, speaker, chip and assigned term code.	863,280 × 4
`chip.txt` Chip ID and its WCS chart coordinates.	330 × 4
`dict.txt` Maps language-specific term codes to recorded transcriptions.	2,364 × 4
`foci-exp.txt` Focal-colour selections, expanded to one row per selected chip.	110,784 × 5
`spkr-lsas.txt` Language, speaker ID, approximate age and sex, where available.	2,619 × 4
`cnum-vhcm-lab-new.txt` Chip-to-Munsell mapping and CIELAB coordinates (L^*, a^*, b^*).	330 × 9

Are richer inventories associated with finer average partitions?



Question 1: outcomes and estimand

The unit of analysis is one language. I fit separate models for two language means:

Mean near-split	Mean compactness
The average across eligible speakers of the fraction of CIELAB-neighbour pairs assigned different names. A larger value indicates finer local lexical differentiation on the fixed neighbour graph.	The average standardised inverse within-term CIELAB dispersion. A larger value indicates that chips assigned the same term lie closer to their category centroid.

The central comparison asks whether recurrent inventory richness is associated with these language means after accounting for eligible-speaker count and, in the expanded model, mean individual term count.

First test: what counts as shared vocabulary richness?

Question. Do languages with richer colour vocabularies partition colour space more finely?

Let ℓ index a language, s a speaker, and t a colour term. Let $\mathcal{S}_\ell$ be the set of eligible speakers in language ℓ, and let $n_\ell = |\mathcal{S}_\ell|$ be their number.

Count recurrent terms across speakers

For each language and term pair, define

$$N_{\ell t} = \#\{s \in \mathcal{S}_\ell : \text{speaker } s \text{ uses } t \text{ on at least one observed chip}\}.$$

The recurrent inventory size is

$$R_\ell = \sum_t \mathbf{1}\{N_{\ell t} \geq 2\}.$$

Here $\mathbf{1}\{A\} = 1$ if statement A is true and 0 otherwise. Thus, a term contributes to R_ℓ only when at least two eligible speakers use it.

Second richness measure: an individual's term count

Let $\mathcal{O}_s$ be the set of colour chips for which speaker s supplied a naming response. For $i \in \mathcal{O}_s$, let Y_{si} be the term assigned by speaker s to chip i.

Individual vocabulary size

$$K_s = \left|\{Y_{si} : i \in \mathcal{O}_s\}\right|.$$

K_s counts distinct terms used by speaker s on observed chips.

Peer estimate for a held-out speaker

For target speaker s in language ℓ, the mean term count among the other eligible speakers is

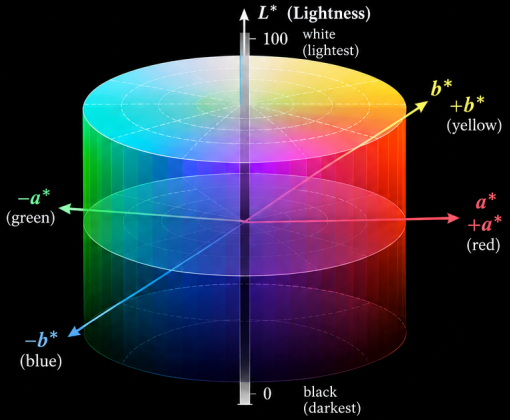
$$\bar{K}_{\ell,-s} = \frac{1}{n_\ell - 1} \sum_{\substack{r \in \mathcal{S}_\ell \\ r \neq s}} K_r.$$

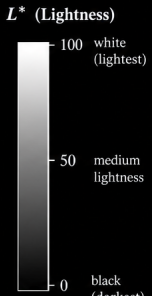
Excluding s prevents their own responses from contributing to this predictor.

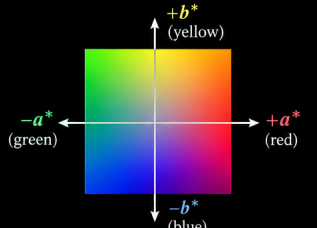
R_ℓ counts recurrent terms. K_s counts terms used by one speaker. $\bar{K}_{\ell,-s}$ averages individual counts over peers.

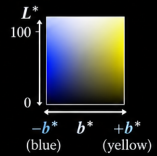
The CIELAB colour space (L^*, a^*, b^*)

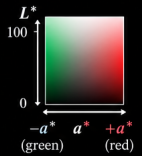
A 3D, approximately perceptually–uniform colour space











Example colours (L^*, a^*, b^*)

$L^* = 70$ $a^* = 30$ $b^* = 10$	$L^* = 60$ $a^* = -20$ $b^* = -40$	$L^* = 80$ $a^* = -10$ $b^* = 60$	$L^* = 30$ $a^* = 20$ $b^* = -30$

Distance in CIELAB

Euclidean distance in (L^*, a^*, b^*) correlates reasonably well with perceived colour difference:

$$\Delta E^*_{ab} = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}$$

Key points

- L^*: lightness (0 = black, 100 = white)
- a^*: green (−) ↔ red (+)
- b^*: blue (−) ↔ yellow (+)
- Approximately perceptually uniform (equal distances ≈ equal perceived differences, though not perfect)

Metric 1a: build the local neighbour graph

Each chip i has CIELAB coordinates $\mathbf{x}_i = (L_i^*, a_i^*, b_i^*)$. For chips i and j, their CIE 1976 Euclidean distance is

$$\Delta E_{76}(i,j) = \sqrt{(L_i^* - L_j^*)^2 + (a_i^* - a_j^*)^2 + (b_i^* - b_j^*)^2}.$$

From distances to unique pairs

Let $\mathcal{N}_3(i)$ be the set of the three chips closest to chip i under this distance. Build

$$\mathcal{E} = \big\{\{i,j\} : j \in \mathcal{N}_3(i) \text{ or } i \in \mathcal{N}_3(j)\big\}.$$

Each element is an **unordered pair**: $\{i,j\} = \{j,i\}$, so the same pair is retained only once even if both chips select each other. The relationship need not be mutual for the pair to be included.

$k = 3$ is a design choice to represent local colour neighbours.

Metric 1b: turn neighbouring pairs into a speaker score

Let $\mathcal{O}_s$ be the set of chips named by speaker s, and let Y_{si} be the term assigned to observed chip i. Retain only neighbour pairs for which both names were observed:

$$\mathcal{E}_s = \{\{i, j\} \in \mathcal{E} : i, j \in \mathcal{O}_s\}.$$

For each observed pair, define

$$D_{sij} = \mathbf{1}\{Y_{si} \neq Y_{sj}\}.$$

Thus $D_{sij} = 1$ if the labels differ and 0 if they agree. The local lexical-resolution score is

$$Q_s^{\text{split}} = \frac{\sum_{\{i,j\} \in \mathcal{E}_s} D_{sij}}{|\mathcal{E}_s|}.$$

Interpretation

Q_s^{split} is the fraction of observed neighbouring pairs that speaker s names differently. A higher value means finer lexical differentiation on this fixed neighbour graph.

The score is undefined if $|\mathcal{E}_s| = 0$ and such a case will be omitted in this metric's analysis.

Metric 2a: naming categories and centroids

For speaker s, let $\mathcal{O}_s$ be the set of chips with an observed name. For each $i \in \mathcal{O}_s$, Y_{si} is the term assigned to chip i, and $\mathbf{x}_i = (L_i^*, a_i^*, b_i^*)$ is its CIELAB coordinate vector.

The chips assigned to a term

For a term t used by speaker s, define the category's chip set

$$I_{st} = \{i \in \mathcal{O}_s : Y_{si} = t\}.$$

Its size is $|I_{st}|$. If $|I_{st}| > 0$, the category's CIELAB centroid is

$$\boldsymbol{\mu}_{st} = \frac{1}{|I_{st}|} \sum_{i \in I_{st}} \mathbf{x}_i.$$

- I_{st} contains exactly the chips speaker s names with term t.
- $\boldsymbol{\mu}_{st}$ is their coordinate-wise mean. It does not need to coincide with a displayed chip.
- An unused term has no centroid for that speaker and is not included in the calculation.

Metric 2b: quantify within-category dispersion

Use $\mathcal{O}_s$, Y_{si}, $\mathbf{x}_i$ and the category centroids $\boldsymbol{\mu}_{st}$ defined on the preceding frame. Let $m_s = |\mathcal{O}_s|$ be the number of chips named by speaker s.

Root-mean-square within-term distance

$$W_s = \sqrt{\frac{1}{m_s} \sum_{i \in \mathcal{O}_s} \left\| \mathbf{x}_i - \boldsymbol{\mu}_{s, Y_{si}} \right\|_2^2}.$$

Here $\| \cdot \|_2$ is Euclidean length in the three CIELAB dimensions. Smaller W_s means the named chips lie closer to their assigned category centroids.

Set $U_s = -W_s$. Let $\bar{U}$ and s_U be the sample mean and standard deviation of U_s across eligible speakers with defined scores. Then

$$Z_s^{\text{compact}} = \frac{U_s - \bar{U}}{s_U}.$$

P.S. U_s has been set to $-W_s$ so that higher values of U_s correspond to greater compactness. Higher Z_s^{compact} means smaller within-term dispersion. If every observed chip has its own term, $W_s = 0$.

Language means: outcome and predictors

Unit of analysis. One row represents one language ℓ. The two primary responses are fitted separately.

$$\bar{Q}_\ell^{\text{split}} = \text{mean near-split score in language } \ell, \qquad \bar{Q}_\ell^{\text{compact}} = \text{mean standardised compactness score.}$$

Predictors

- $\log(n_\ell)$ adjusts for the number of eligible speakers.
- R_ℓ counts recurrent colour terms used by at least two eligible speakers.
- $\bar{K}_\ell$ is the mean number of distinct colour terms used by an eligible speaker.

The response is a language mean. The predictor $\bar{K}_\ell$ is derived from the same naming task and is therefore a concurrent explanatory variable rather than a pre-response predictor.

Ridge and lasso: predictor sequence for language means

Each outcome is fitted separately using one row per language. The three predictor sets are

$$\begin{aligned} M_0: &\quad \log(n_\ell), \\ M_1: &\quad \log(n_\ell) + R_\ell, \\ M_2: &\quad \log(n_\ell) + R_\ell + \bar{K}_\ell. \end{aligned}$$

Here n_ℓ is the eligible-speaker count, R_ℓ is recurrent inventory richness, and $\bar{K}_\ell$ is mean individual term count.

The two penalties

Ridge

$$\mathrm{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2$$

Shrinks correlated coefficients towards zero.

Lasso

$$\mathrm{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|$$

Can set some coefficients exactly to zero.

Lambda is selected by five-fold inner cross-validation inside each training set. Performance is assessed over five repeats of outer five-fold cross-validation, with all 109 languages predicted in each repeat. The supplied CV summary contains M1 and M2 scores only. The one-predictor M_0 ridge/lasso fits returned errors, so no M_0 RMSE is available and no M_1-versus-M_0 CV conclusion is made.

Nested cross-validation: what is held out?

Inner loop: choose the penalty

Within each outer training set, five-fold cross-validation selects λ for ridge or lasso. The outer test fold is not used to tune the penalty.

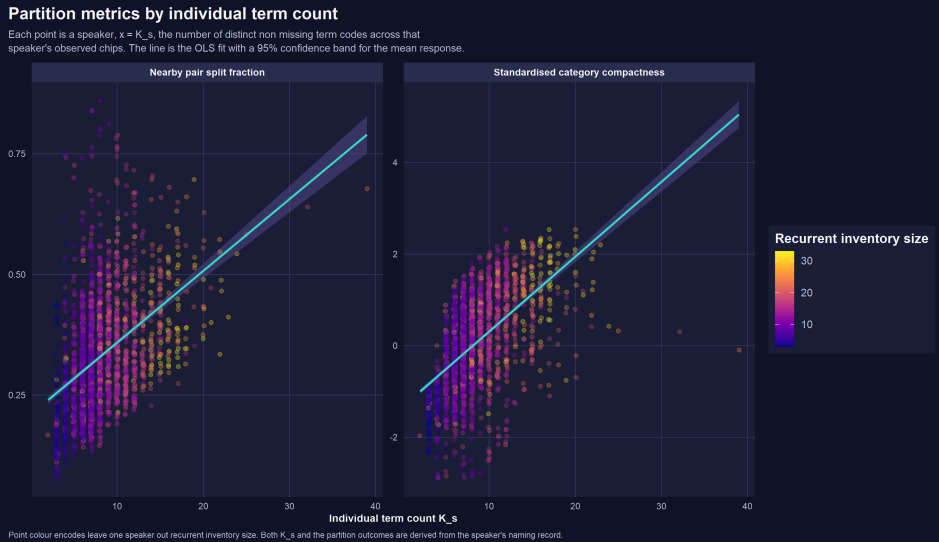
Outer loop: assess prediction

We hold out about one fifth of the 109 languages, fit on the other four fifths, and predict the held-out languages' mean outcomes. Then we rotate through all five folds and repeat the outer split five times.

What this evaluates

The target is the language-level mean near-split or compactness score. The held-out unit is a language, not a speaker. RMSE compares predicted and observed language means across outer folds.

This is held-out-language prediction conditional on observed language-level predictors. R_ℓ and $\bar{K}_\ell$ are computed from that language's naming data. The supplied one-predictor M_0 fits returned errors and hence valid CV scores are available for M_1 and M_2 only.



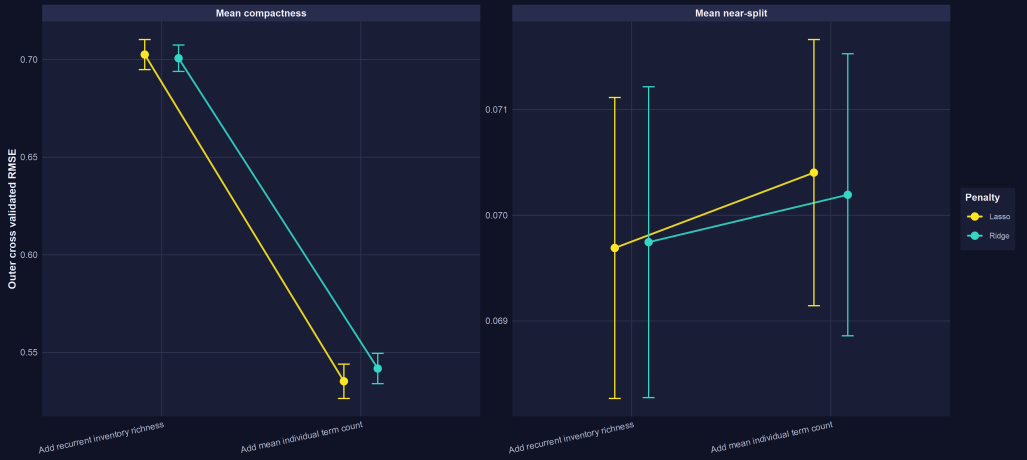
Each point is a language. Mean individual term count is strongly associated with compactness. Since it is derived from the same naming task, it is a concurrent explanatory variable rather than a pre-response predictor.

Results: what improves held-out prediction?

Outcome	Penalty	M1 RMSE (SD)	M2 RMSE (SD)	ΔRMSE	R^2_{pred} M1 → M2
Near-split	Lasso	0.06969 (0.00142)	0.07040 (0.00126)	+0.00071	0.238 → 0.222
Near-split	Ridge	0.06975 (0.00147)	0.07019 (0.00133)	+0.00045	0.237 → 0.227
Compactness	Lasso	0.70256 (0.00769)	0.53523 (0.00879)	−0.16734	0.187 → 0.528
Compactness	Ridge	0.70069 (0.00677)	0.54177 (0.00776)	−0.15892	0.192 → 0.517

- For near-split, adding mean individual term count in M2 slightly increases RMSE under both penalties. It does not improve held-out prediction over M1.
- For compactness, M2 reduces RMSE by about 0.159 to 0.167 and raises predictive R^2 from about 0.19 to 0.52 to 0.53.

Each repeat predicts all 109 languages. Lower RMSE is better. Parentheses give the standard deviation across the five outer-CV repeats.



Points show mean outer-fold RMSE over five repeated nested five-fold cross-validation runs. Bars show one standard deviation across repeats. M1 contains log eligible-speaker count and recurrent richness. M2 additionally contains mean individual term count. Lower RMSE is better. The plot compares M1 with M2 because M0 returned no valid CV scores.

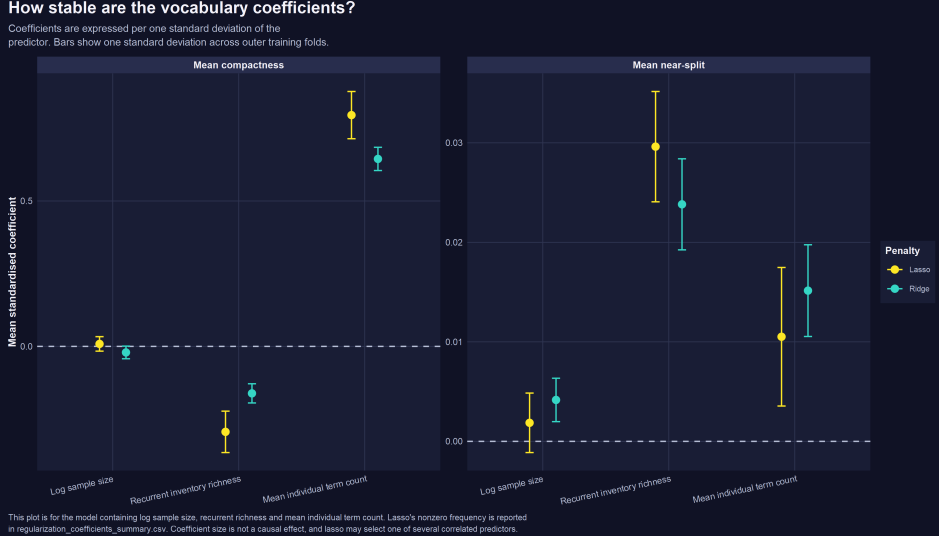
Reading the regularised vocabulary coefficients

The table gives mean coefficients from the M2 models, expressed per one standard deviation of each predictor across the outer training folds.

Outcome	Penalty	Recurrent richness R_ℓ	Mean terms $\bar{K}_\ell$
Near-split	Ridge	$+0.0238$	$+0.0151$
Near-split	Lasso	$+0.0296$	$+0.0105$
Compactness	Ridge	-0.1616	$+0.6447$
Compactness	Lasso	-0.2939	$+0.7953$

- For near-split, both vocabulary coefficients remain positive under both penalties.
- For compactness, mean individual term count has a large positive coefficient, while the richness coefficient turns negative after conditioning on mean term count.
- The two language-level vocabulary measures are strongly correlated ($r = 0.831$). The conditional sign reversal is therefore a shared-information issue, not evidence that richer inventories have a negative marginal association with compactness.

Lasso retains mean term count in 84% of near-split outer fits and 100% of compactness outer fits.

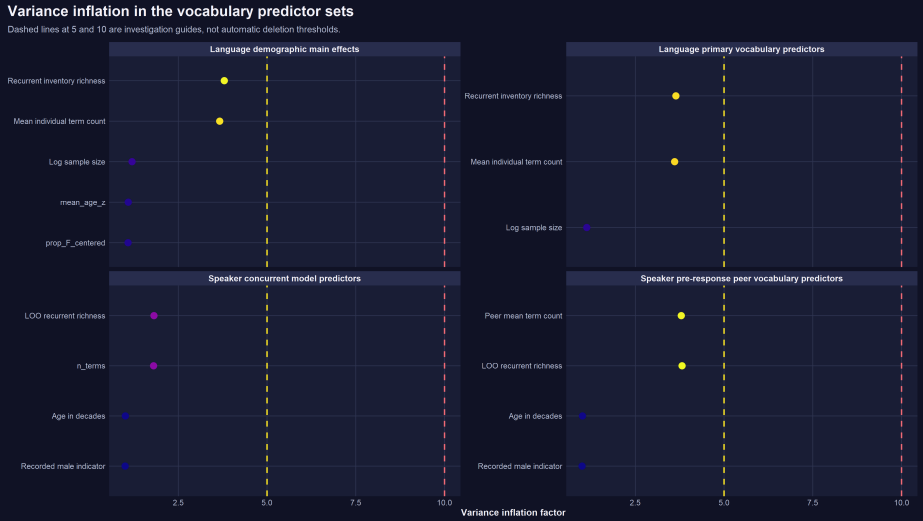


M2 coefficients are expressed per one standard deviation of each predictor. Bars show one standard deviation across outer training folds, not confidence intervals. The compactness richness coefficient is conditional on mean term count.

The vocabulary measures share information

Predictor pair	Pearson r	VIF (richness)	VIF (mean terms)
Language: R_ℓ and $\bar{K}_\ell$	0.831	3.64	3.61
Speaker peers: $R_{\ell,-s}$ and $\bar{K}_{\ell,-s}$	0.859	3.82	3.80

The simple vocabulary predictor sets have VIFs below the common value of 5 used as an investigation guide. However, the high pairwise correlations show that recurrent richness and mean term count carry overlapping information. These values do not describe the interaction-expanded models. Their separate diagnostics are higher: individual near-split VIF reaches 49.03. Interaction coefficients therefore need particular caution.



Displayed VIFs cover the simple language and speaker predictor sets. All are below 5. The interaction-expanded designs have higher VIFs, reaching 49.03 in the individual near-split interaction model.



Recurrent inventory richness and mean individual term count correlate 0.83 across languages. Their leave-one-speaker-out peer versions correlate 0.86. Pairwise correlation and VIF describe related but different aspects of predictor overlap.

From the first question to the next

What we learnt

- Adding mean term count does not improve near-split prediction over the sample-size-plus-richness M1 model.
- The same addition substantially improves compactness prediction.
- Recurrent richness and mean individual term count overlap, so compactness coefficients are conditional on their shared information.

The next question

An average might improve while a subset of speakers remains at the bottom, or richness might be related to both tails.

So we move from mean outcomes to the distribution: who appears in the low and high deciles?

Are richer languages represented differently at the extremes?



Working assumptions for the OLS analyses

For an OLS model at its relevant observation unit i,

$$Y_i = \mathbf{x}_i^{\mathsf{T}} \beta + \varepsilon_i.$$

The unit is a language in the tail-prevalence and regional-mean regressions, and a speaker in the demographic and individual-score regressions.

- **Mean specification and exogeneity:** the conditional mean is adequately represented and $E(\varepsilon_i \mid \mathbf{X}) = 0$.
- **No perfect multicollinearity:** no predictor is an exact linear combination of the others. High correlation can still inflate standard errors.
- **Independence:** errors should be independent across the model's observational units. Speakers within a language may still share unobserved influences, which is a limitation of the individual-level fits.
- **Homoscedasticity:** $\mathrm{Var}(\varepsilon_i \mid \mathbf{X}) = \sigma^2$. A changing residual spread can invalidate conventional standard errors.
- **Normality:** mainly supports exact small-sample t and F inference; it is not required for OLS unbiasedness under exogeneity.

The grouped-binomial GLMs have a separate binomial response assumption.

How the OLS and GLM sections compare models

Let n be the number of observations, k the number of parameters counted by the likelihood criterion, and $\widehat{\mathcal{L}}$ the maximised likelihood. The information criteria are

$$AIC = -2\log\widehat{\mathcal{L}} + 2k, \qquad BIC = -2\log\widehat{\mathcal{L}} + k\log n.$$

Lower AIC/BIC is preferred only for models fitted to the same response and observations. BIC generally penalises extra parameters more strongly when n is large.

For observed responses y_i, let $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$, and let $\hat{y}_i$ be the fitted value. Then

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

For p predictors excluding the intercept,

$$R^2_{\mathrm{adj}} = 1 - \frac{n-1}{n-p-1}(1 - R^2).$$

Turning speaker scores into tail proportions

Choose a speaker-level metric Q_s (for example, Q_s^{split} or Z_s^{compact}). Let $q_{.10}$ and $q_{.90}$ be its empirical 10th and 90th percentiles among eligible speakers with a defined metric.

- **Speaker-level tail indicators**

  Define

  $$L_s = \mathbf{1}\{Q_s \le q_{.10}\}, \qquad H_s = \mathbf{1}\{Q_s \ge q_{.90}\}.$$

  $L_s = 1$ denotes a bottom-decile score, while $H_s = 1$ denotes a top-decile score.

- **Language-level prevalence**

  For language ℓ, let $\mathcal{S}_\ell$ be the speakers with a defined metric and $n_\ell = |\mathcal{S}_\ell|$. Define the numbers of lower-tail and upper-tail speakers as

  $$x_{\ell,L} = \sum_{s \in \mathcal{S}_\ell} L_s, \qquad x_{\ell,H} = \sum_{s \in \mathcal{S}_\ell} H_s.$$

  The corresponding language-level proportions are

  $$p_{\ell,L} = \frac{x_{\ell,L}}{n_\ell}, \qquad p_{\ell,H} = \frac{x_{\ell,H}}{n_\ell}.$$

Tail proportions: transformed linear model

For one selected tail, let x_ℓ be the number of speakers in that tail and n_ℓ the number eligible in language ℓ. The observed proportion is $\widehat{p}_\ell = x_\ell / n_\ell$.

Arcsine-square-root transformation

$$Z_\ell = \arcsin \sqrt{\widehat{p}_\ell}.$$

The language-level regression is

$$Z_\ell = \beta_0 + \beta_1 \log(n_\ell) + \beta_2 R_\ell + \varepsilon_\ell,$$

where R_ℓ is recurrent inventory richness, β_j are regression coefficients, and ε_ℓ is the language-level error.

Our response is a proportion of speakers in a particular tail. Proportions have a variance that depends on both the underlying proportion and the number of speakers observed. In particular, proportions around $\frac{1}{2}$ are more variable than proportions close to 0 or 1. We therefore apply the arcsine-square-root transformation. This transformation approximately cancels the dependence of the variance on the underlying proportion, making the response more suitable for linear regression.

Tail proportions: grouped-binomial GLM sensitivity

Let $X_\ell = x_\ell$ be the number of speakers in the chosen tail, n_ℓ the eligible denominator, and π_ℓ the modelled probability that an eligible speaker in language ℓ falls in that tail.

$$X_\ell \sim \text{Binomial}(n_\ell, \pi_\ell), \qquad \log \frac{\pi_\ell}{1-\pi_\ell} = \mathbf{z}_\ell^\top \beta.$$

Here $\mathbf{z}_\ell$ is the vector of predictors (including an intercept), and β is the coefficient vector. The logit link keeps fitted probabilities between 0 and 1.

- We model the observed integer success and failure counts directly, rather than approximating them with fractional prior weights.
- The binomial model assumes conditionally independent Bernoulli outcomes with a common probability within language.
- AIC/BIC comparisons use the same counts and analysis sample.

Baseline and added ladders: language-level tail prevalence

The analysis is fitted separately for near-split and compactness, and separately for the lower and upper tail. For language ℓ, $\bar{A}_\ell$ is mean recorded age in decades among speakers with age data, and F_ℓ is the proportion recorded female among speakers with sex data. These summaries are used in the grouped-binomial GLM sensitivity sequence.

A. Arcsine square root OLS

Response: $Z_\ell = \arcsin\sqrt{x_\ell / n_\ell}$.

1. TL_0: baseline with $\log(n_\ell)$.
2. TL_1: add recurrent inventory richness R_ℓ.
3. TL_2: add mean individual term count $\bar{K}_\ell$.

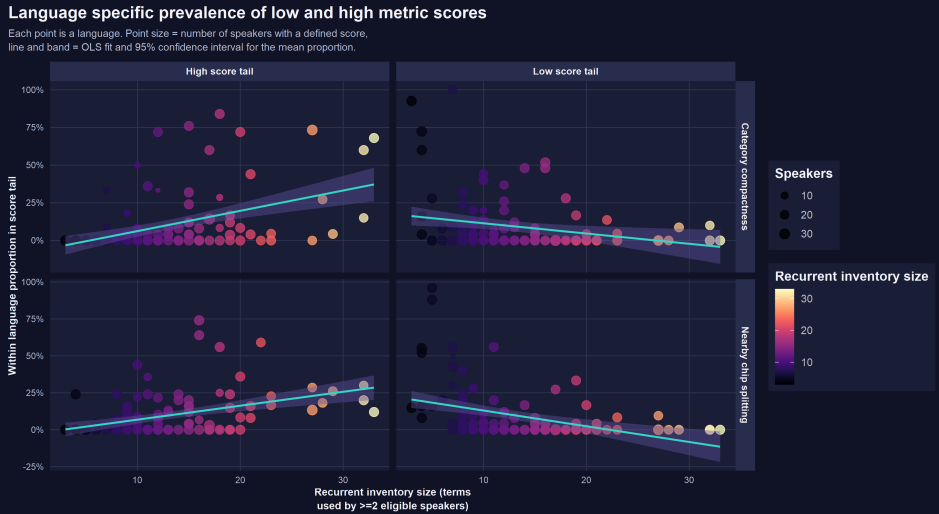
The result table reports the richness and mean term count steps.

B. Grouped binomial GLM sensitivity

Response: $X_\ell \sim \text{Binomial}(n_\ell, \pi_\ell)$ with a logit link.

1. G_0: intercept only.
2. G_1: add recurrent inventory richness R_ℓ.
3. G_2: add mean individual term count $\bar{K}_\ell$.
4. G_3: add $\bar{A}_\ell$ and F_ℓ.
5. G_4: add $\bar{A}_\ell R_\ell$, $F_\ell R_\ell$, and $\bar{A}_\ell F_\ell$.
6. G_5: add $\bar{A}_\ell F_\ell R_\ell$.

The AIC and BIC plots show the penalty comparison for this sequence.



Each dot is a language. The vertical axis is the observed fraction of speakers in the selected lower or upper tail. Panels separate near-split and compactness, then lower and upper tails. This is descriptive prevalence before the adjusted regressions.

Results: language-level tail prevalence

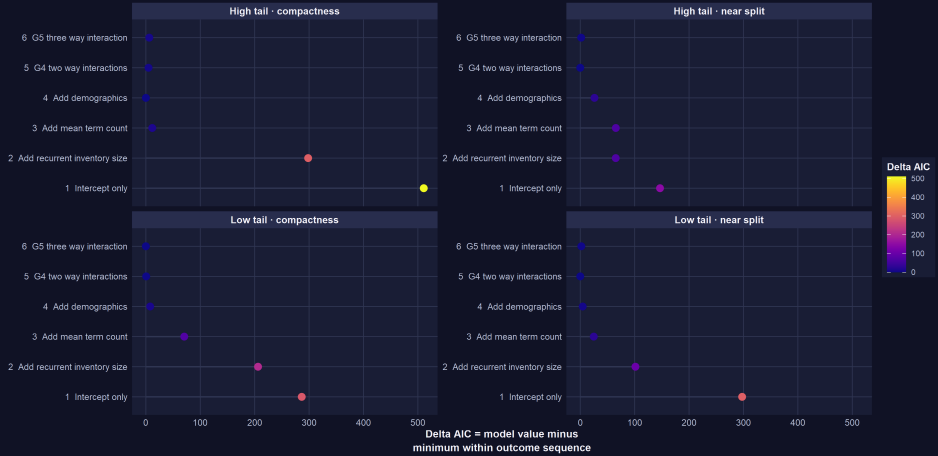
Here $\hat{\beta}_R$ is the estimated richness coefficient, $p(\hat{\beta}_R)$ is its nominal coefficient-test p-value, and ΔBIC is current BIC minus the previous model's BIC. TL_1 adjusts for $\log(n_\ell)$; TL_2 additionally adjusts for mean individual term count $\bar{K}_\ell$.

Outcome/tail	Model	Adj. R^2	$\hat{\beta}_R$	$p(\hat{\beta}_R)$	BIC change
Near-split low	TL_1: add richness	0.185	-0.0196	0.000002	-18.97
Near-split low	TL_2: add mean terms	0.273	$+0.0021$	0.758	-8.69
Near-split high	TL_1: add richness	0.243	$+0.0182$	$< 10^{-6}$	-22.68
Near-split high	TL_2: add mean terms	0.246	$+0.0248$	0.00014	$+3.12$
Compactness low	TL_1: add richness	0.066	-0.0117	0.0078	-2.64
Compactness low	TL_2: add mean terms	0.207	$+0.0166$	0.029	-14.00
Compactness high	TL_1: add richness	0.232	$+0.0229$	$< 10^{-7}$	-25.40
Compactness high	TL_2: add mean terms	0.504	-0.0161	0.0086	-43.48

All TL_2 richness coefficients are conditional on mean individual term count. Negative ΔBIC favours the expanded model; positive ΔBIC favours the previous model.

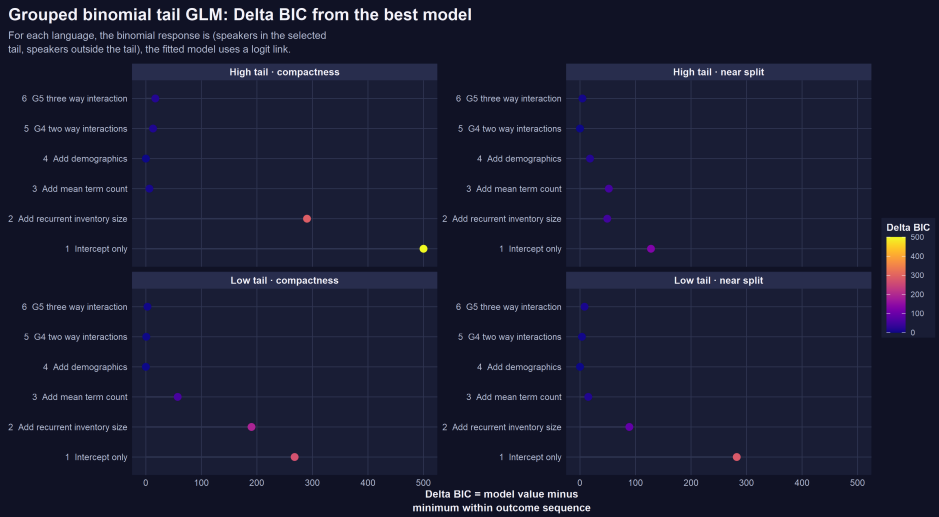
Grouped binomial tail GLM: Delta AIC from the best model

For each language, the binomial response is (speakers in the selected tail, speakers outside the tail), the fitted model uses a logit link.



Within each sequence, Delta AIC = AIC(model) minus minimum AIC. AIC = minus 2 log(L) + 2k, BIC = minus 2 log(L) + k log(n), where L is the maximised likelihood, k is the parameter count and n is the fitted sample size.

Each point is AIC minus the minimum within an outcome. G4 has the lowest AIC for both near-split tails. G5 is only slightly lower than G4 for compactness lower tail, while G3 is lowest for compactness upper tail. These are in-sample criteria, not held-out prediction.



Within each sequence, Delta BIC = BIC(model) minus minimum BIC. AIC = minus 2 log(L) + 2k. BIC = minus 2 log(L) + k log(n), where L is the maximised likelihood, k is the parameter count and n is the fitted sample size.

Each point is BIC minus the minimum within an outcome. G3 has the lowest BIC for both compactness tails and the near-split lower tail. G4 has the lowest BIC for the near-split upper tail. G5 is not the BIC winner for any outcome.

Interpretation: language-level tail prevalence

- **Near-split, lower tail.** The richness coefficient changes from -0.0196 ($p = 0.000002$) in TL_1 to $+0.0021$ ($p = 0.758$) in TL_2. The initial negative association is no longer evident after adjusting for mean individual term count.
- **Near-split, upper tail.** The richness coefficient remains positive, changing from $+0.0182$ to $+0.0248$. However, adding mean term count raises BIC by 3.12, so that extra term does not improve the penalised comparison.
- **Compactness tails.** The lower-tail coefficient changes from -0.0117 to $+0.0166$, while the upper-tail coefficient reverses from $+0.0229$ to -0.0161 after adding mean individual term count.
- **Interpretation.** The tails are not mirror images, and the richness association depends on whether mean term count is included. Since both vocabulary measures come from the same naming task, these are conditional associations, not causal effects.

Do demographic effects vary, and does richness operate differently across groups?



Question 3 has two distinct parts

A. Demographic heterogeneity	B. Richness moderation
Do age and recorded-sex associations have a common value across languages, or do the age slopes and sex contrasts vary by language?	Does the association between recurrent vocabulary richness and an individual's partition score change with age or recorded sex?
Comparison: common effects versus language-specific effects.	**Comparison:** main-effects model versus two-way and three-way interactions.

Both parts use speaker-level outcomes, but they test different hypotheses. A language-specific age slope is not the same estimand as an age-by-richness interaction.

A. Do demographic effects vary across languages?

- **Age differences**
  - Does age have the same association with the speaker's colour-partition measure in every language?
  - Or do age slopes differ across languages?
- **Sex differences**
  - Is the difference between the recorded female and male groups similar across languages?
  - Or does its size and direction vary by language?
- **Why compare the models?**
  - A common slope uses fewer parameters and gives a simpler model.
  - Language-specific slopes add parameters and must improve model fit enough to justify the added complexity.
- **Statistical target**
  - Compare common-effect and language-specific-effect regressions using the same observations.
  - Assess whether demographic associations vary across languages, not whether age or sex causally changes colour perception.

Age slopes: common or language-specific?

Let Y_s be a continuous speaker-level outcome, $A_s = \text{age}_s/10$ age in decades, $S_s = 1$ for recorded male and $S_s = 0$ for recorded female, and $\ell(s)$ the language of speaker s. The β terms are regression coefficients, α_ℓ is a language-specific intercept, and ε_s is the residual.

Baseline model M_{A0}

$$Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \alpha_{\ell(s)} + \varepsilon_s.$$

All languages share the same age slope β_A.

Expanded model M_{A1}

$$Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \alpha_{\ell(s)} + \delta_{\ell(s)} A_s + \varepsilon_s.$$

Adds language-specific deviations δ_ℓ to the age slope. The reference language has $\delta_{\ell_0} = 0$.

With reference coding, $\alpha_{\ell_0} = 0$. The nested F-test assesses whether the added age-slope deviations improve the model jointly.

Sex differences: common or language-specific?

Use the same outcome Y_s, age predictor A_s, language indicator $\ell(s)$ and sex indicator S_s. The coefficient β_S represents the common conditional difference between recorded male and female groups. The terms γ_ℓ allow that difference to vary across languages.

Baseline model M_{S0}

$$Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \alpha_{\ell(s)} + \varepsilon_s.$$

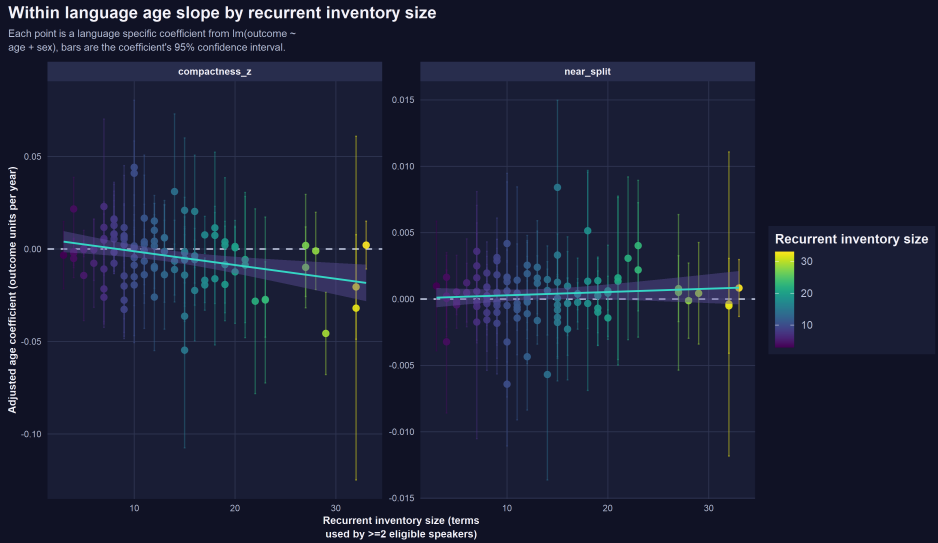
Every language shares the same conditional sex difference β_S.

Expanded model M_{S1}

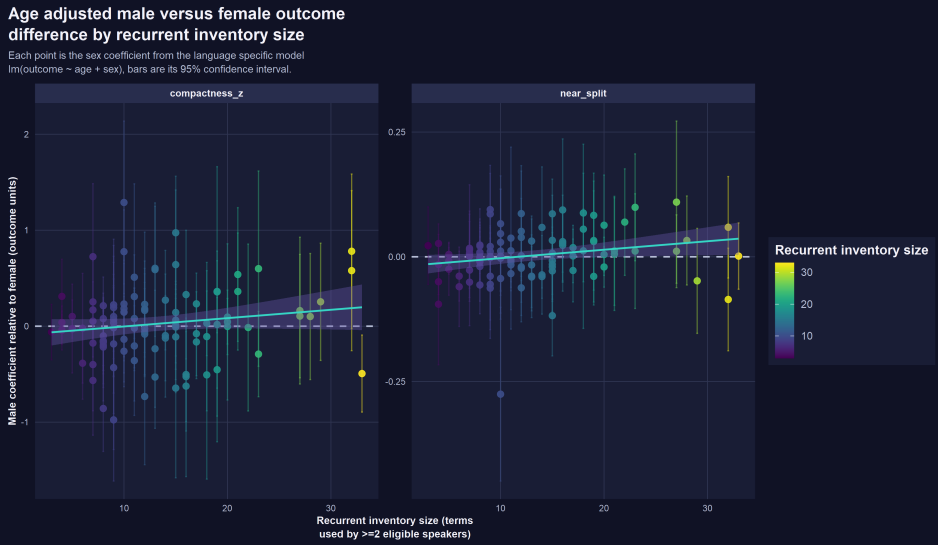
$$Y_s = \beta_0 + \beta_A A_s + \alpha_{\ell(s)} + \beta_S S_s + \gamma_{\ell(s)} S_s + \varepsilon_s.$$

Adds sex-by-language terms γ_ℓ, allowing the conditional sex difference to vary across languages.

With reference coding, $\alpha_{\ell_0} = 0$ and $\gamma_{\ell_0} = 0$. The sex-by-language block is tested jointly. Interpret the F-test alongside AIC, BIC and PRESS to account for the added model complexity.



Each point is a language-specific age slope from the within-language regression. The horizontal axis is recurrent richness and the vertical axis is the estimated outcome change per decade. Error bars show uncertainty; visual spread alone does not establish heterogeneous slopes.



Each point is a language-specific recorded-sex contrast plotted against recurrent richness. Error bars show uncertainty. These estimates describe variation across languages and do not by themselves establish heterogeneity.



These panels show descriptive speaker-level outcome patterns by age and recorded sex. They motivate, but do not replace, the common-effect versus language-specific-effect model comparisons.

Heterogeneity results: the test and the penalty tell different stories

Outcome	Extra block	Unadj. R^2 (base $\rightarrow$ expanded)	ΔAIC	ΔBIC	ΔPRESS
Near-split	age slopes by language	.479 $\rightarrow$ .509	+58.7	+627.6	+0.713
Near-split	sex effects by language	.479 $\rightarrow$ .519	+9.3	+578.2	+0.192
Compactness	age slopes by language	.630 $\rightarrow$ .660	+6.5	+575.4	+16.26
Compactness	sex effects by language	.630 $\rightarrow$ .658	+17.0	+585.9	+14.67

These are ordinary R^2 values from the separate common-slope versus language-specific-slope models, which include language-specific intercepts.

- **Nested F-tests detect demographic heterogeneity.** Allowing age slopes to vary across languages is significant for near-split ($p = 0.0198$) and highly significant for compactness ($p = 1.60 \times 10^{-6}$). Allowing recorded-sex differences to vary across languages is also highly significant for both near-split ($p = 2.94 \times 10^{-6}$) and compactness ($p = 1.50 \times 10^{-5}$). These are joint tests of the added language-specific effects, not tests of each language coefficient separately.
- **Statistical significance does not justify the expansion.** Each expanded model adds 99 parameters, increasing the parameter count from 102 to 201. Despite the significant F-tests, all four expansions increase AIC, BIC and PRESS. Thus, the additional language-specific effects improve in-sample fit but do not justify their complexity under the penalised criteria or improve leave-one-out prediction.

B. Does richness moderate demographic associations?

Age moderation

Is the association between language richness and a speaker's continuous partition measure different for younger and older speakers, after accounting for the age association itself?

Joint moderation

Does the age-by-richness relationship itself vary by sex? This is the role of the three-way age-by-sex-by-richness term.

Sex moderation

Does the richness association differ by recorded sex, rather than merely showing an overall difference between the sex groups?

Decision question

Do these interaction terms improve fit or prediction enough to justify their added complexity, or does a simpler common-effect model suffice?

Shared richness for a held-out speaker

For a target speaker s in language ℓ, define the peer set

$$\mathcal{S}_{\ell,-s} = \mathcal{S}_\ell \setminus \{s\}.$$

For each term t, count how many of those other speakers use it on at least one observed chip:

$$N_{\ell t,-s} = \#\{r \in \mathcal{S}_{\ell,-s} : \exists i \in \mathcal{O}_r \text{ with } Y_{ri} = t\}.$$

The leave-one-speaker-out recurrent inventory is

$$R_{\ell,-s} = \sum_t \mathbf{1}\{N_{\ell t,-s} \geq 2\}.$$

Why exclude the target speaker?

When $R_{\ell,-s}$ predicts speaker s's outcome, that speaker's own naming responses do not help construct the predictor. A term counts only if at least two of the *other* eligible speakers use it.

$R_{\ell,-s}$ counts the number of recurrent terms among peers and $\bar{K}_{\ell,-s}$ counts the average number of distinct terms peers use.

Outcome, predictors and interaction ladder

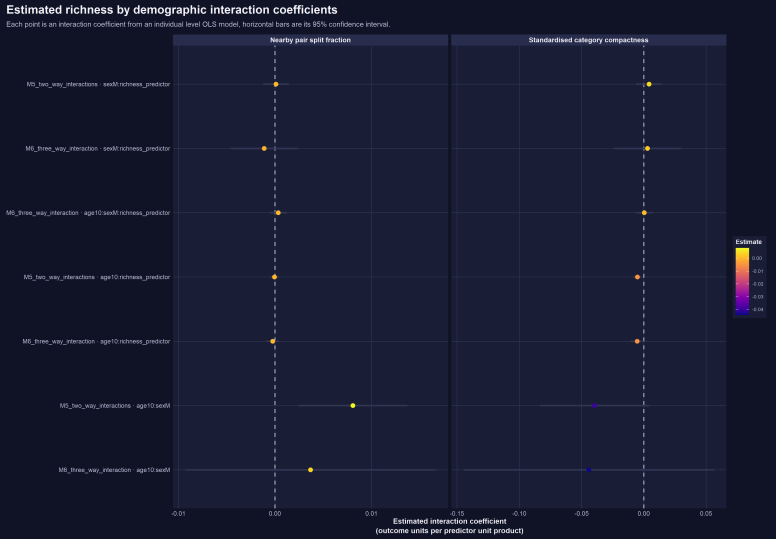
Let Y_s be near-split or compactness, $A_s = \mathrm{age}_s/10$, $S_s = 1$ for recorded male and 0 for recorded female, $R_s = R_{\ell(s),-s}$ be leave-one-speaker-out language richness, and K_s be the speaker's own number of distinct terms.

Exact sequence behind the interaction-penalty results

$$IM_0: \quad Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \beta_R R_s + \beta_K K_s + \varepsilon_s,$$

$$IM_1: \quad IM_0 \text{ plus } \beta_{AR} A_s R_s + \beta_{SR} S_s R_s + \beta_{AS} A_s S_s,$$

$$IM_2: \quad IM_1 \text{ plus } \beta_{ASR} A_s S_s R_s.$$



Intervals show estimated two-way and three-way interaction coefficients for age, recorded sex and leave-one-speaker-out richness. Intervals crossing zero indicate uncertainty. The model-comparison table determines whether the extra terms justify their complexity.

Moderation results: added interactions rarely pay for themselves

Outcome	Model step	ΔAIC	ΔBIC	ΔPRESS	Reading
Near-split	IM_1 vs IM_0: add two-way interactions	-2.14	$+15.15$	-0.027	BIC rejects
Near-split	IM_2 vs IM_1: add three-way interaction	$+1.45$	$+7.22$	$+0.015$	reject
Compactness	IM_1 vs IM_0: add two-way interactions	-5.72	$+11.57$	-4.18	trade-off
Compactness	IM_2 vs IM_1: add three-way interaction	$+1.99$	$+7.75$	$+1.22$	reject

- BIC penalises each added parameter by $\log n$, about 7.76 when $n = 2355$. The interaction block adds several terms.
- Two-way terms can slightly reduce PRESS or AIC, but BIC rises and three-way terms make prediction worse.
- This table compares the individual-level IM_0 to IM_2 interaction sequence. The subgroup-gap plots above describe language-level contrasts and are not tested by this table.

Where in the sampled colour space are the associations strongest?



Where in the sampled colour space do the patterns occur?

Localising the richness association

Is the relationship between vocabulary richness and the probability of giving different names to nearby chips similar across all sampled colour regions, or is it stronger in some regions than others?

Age and sex by region

Do age-related differences or recorded-sex differences in naming change across regions of the colour chart, even if an overall demographic association is weak?

Interpretation of a regional effect

A regional difference can reveal where the language-level association is concentrated. It does not establish a universal boundary or a perceptual threshold.

How the regions are defined

Regions are operational groups of the sampled chip pairs based on the chart's hue/chroma coordinates. They are not equal-volume regions of all visible colour space.

Region-specific questions and operational regions

Spatial richness. **Does the richness association differ across sampled colour regions?**

Regional demographics. **Do age/sex differences in naming vary by region?**

Let r index a predefined colour region and $\mathcal{E}_r$ be the unique neighbour pairs assigned to region r. Using Y_{si} for speaker s's name for chip i, define

$$D_{sij} = \mathbf{1}\{Y_{si} \neq Y_{sj}\}.$$

Let $\mathcal{E}_{sr} = \{\{i,j\} \in \mathcal{E}_r : i,j \in \mathcal{O}_s\}$ be the pairs in region r for which speaker s named both chips. The regional split rate is

$$Q_{sr} = \frac{1}{|\mathcal{E}_{sr}|} \sum_{\{i,j\} \in \mathcal{E}_{sr}} D_{sij}.$$

Achromatic, achromatic edge, red to orange, yellow to green, green to cyan, blue to violet, purple to red, and cross-band. These group the sampled pairs using mapped hue and chroma coordinates.

Outcome, predictors and model ladder: regional richness

For speaker s and colour region r, Q_{sr} is the fraction of observed neighbouring chip pairs in that region that receive different labels. For language ℓ, the response is the mean regional score

$$\bar{Q}_{\ell r} = \frac{1}{m_{\ell r}} \sum_{s \in \mathcal{S}_{\ell r}} Q_{sr}.$$

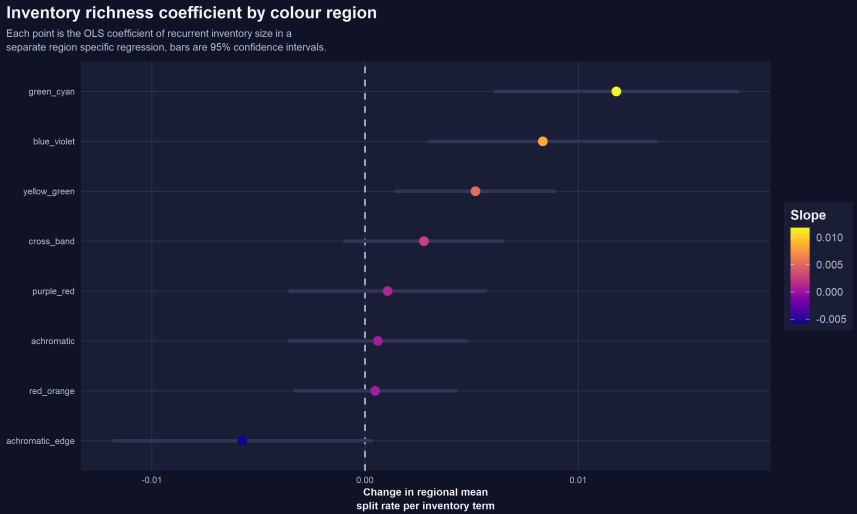
The model is fitted separately for each colour region, with one row per language.

Predictors and added ladder

$$\bar{Q}_{\ell r} = \beta_{0r} + \beta_{nr} \log(n_\ell) + \beta_{Rr} R_\ell + \beta_{Kr} \bar{K}_\ell + \varepsilon_{\ell r}.$$

1. RG_0: intercept only.
2. RG_1: add log eligible speaker count $\log(n_\ell)$.
3. RG_2: add recurrent inventory richness R_ℓ.
4. RG_3: add mean individual term count $\bar{K}_\ell$.

The reported richness coefficient is from RG_3, conditional on sample size and mean term count. The RG labels refer only to this regional OLS sequence.



Each point is the conditional richness coefficient from a separate language-level regression for one sampled colour region. Intervals are 95% confidence intervals. Green-to-cyan and blue-to-violet have the largest positive estimates.

Results: richness slopes across colour regions

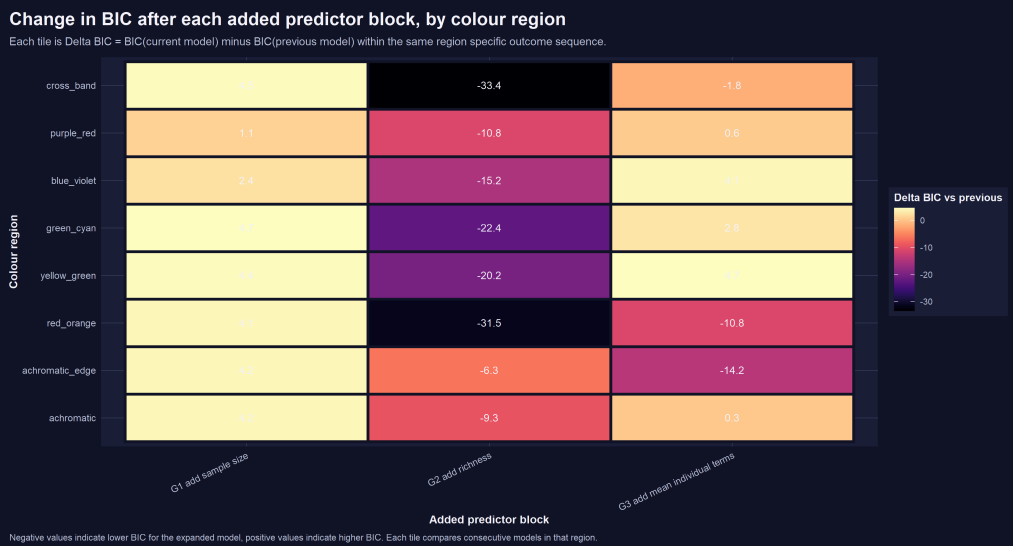
For each region r, $\hat{\beta}_{R,r}$ is the estimated inventory-richness coefficient in the regional OLS model defined on the preceding slide. SE_r is its standard error, and the displayed interval is its 95% confidence interval. The nominal p_r tests $H_0 : \beta_{R,r} = 0$ in that model.

Region r	$\hat{\beta}_{R,r}$	SE_r	95% CI	Nominal p_r
Achromatic	0.0006	0.0021	$[-.0036, .0048]$	.779
Achromatic edge	-0.0058	0.0031	$[-.0119, .0004]$	.065
Red to orange	0.0005	0.0019	$[-.0034, .0043]$	.805
Yellow to green	0.0052	0.0019	$[.0014, .0090]$	.0077
Green to cyan	0.0118	0.0029	$[.0060, .0176]$	.00010
Blue to violet	0.0083	0.0027	$[.0030, .0137]$	.0027
Purple to red	0.0011	0.0023	$[-.0036, .0057]$	.653
Cross-band	0.0028	0.0019	$[-.0010, .0065]$	.149

Row colours identify the colour-space regions. The p_r values are unadjusted for the eight regional tests, so multiple testing should be considered when interpreting significance.

Interpreting the regional richness results

- The estimated conditional richness coefficient is largest in green to cyan (0.0118, 95% CI $[0.0060, 0.0176]$) and next largest in blue to violet (0.0083, 95% CI $[0.0030, 0.0137]$).
- Eight regions were tested. With a simple Bonferroni threshold $0.05/8 = 0.00625$, green to cyan ($p = 0.00010$) and blue to violet ($p = 0.0027$) remain below the threshold. Yellow to green ($p = 0.0077$) does not.
- The regional pattern is suggestive, especially for green to cyan and blue to violet, but should remain exploratory until the regions and contrasts are pre-specified and independently checked.



Each tile is the BIC change from the preceding regional model. Negative values favour the added predictor block. Adding richness lowers BIC in all eight regions; the contribution of mean term count is region-dependent.

Regional demographic contrasts: a separate descriptive analysis

Different target

The regional richness models predict language-level mean near-split scores from vocabulary predictors. The next analysis instead calculates age, sex and joint group gaps within each colour region. These contrasts are descriptive and are not additional predictors in the richness ladder.

What the regional demographic plots estimate

For each language ℓ and colour region r, let $\bar{Q}_{\ell r,g}$ be the mean regional split rate among speakers in demographic group g. The plotted quantities are group contrasts, not extra predictors in the regional richness regression:

$$G_{\ell r}^{\text{age}} = \bar{Q}_{\ell r,O} - \bar{Q}_{\ell r,Y}, \qquad G_{\ell r}^{\text{sex}} = \bar{Q}_{\ell r,M} - \bar{Q}_{\ell r,F}.$$

The joint age-by-sex contrast is a difference-in-differences:

$$G_{\ell r}^{\text{age}\times\text{sex}} = (\bar{Q}_{\ell r,O,M} - \bar{Q}_{\ell r,O,F}) - (\bar{Q}_{\ell r,Y,M} - \bar{Q}_{\ell r,Y,F}).$$

- Each point in the contrast plot represents a language-level regional contrast.
- The BIC heatmap compares the RG_0 through RG_3 richness model sequence defined earlier, one region at a time.



Points show language-level age, recorded-sex and joint subgroup contrasts in regional split rates. Values near zero indicate small average differences. Scattered estimates and wide intervals signal uncertainty rather than a stable regional demographic effect.

Results: demographic differences across regions

- **Age differences.** The largest older minus younger gap occurs in yellow to green (0.0125, or 1.25 percentage points). Its descriptive interval is $[0.0008, 0.0243]$. The other seven regional intervals cross zero.
- **Sex differences.** The largest positive male minus female gap occurs in red to orange (0.0136, or 1.36 percentage points), with descriptive interval $[0.0021, 0.0251]$. The other seven regional intervals cross zero.
- **Age by sex differences.** The difference in male minus female gaps between older and younger speakers is largest in yellow to green (0.0450), followed by blue to violet (0.0427), cross-band (0.0301) and red to orange (0.0294). The plotted intervals for these four regions lie above zero.
- The age, sex and joint contrasts use 94, 101 and 48 languages respectively. The plotted intervals are means plus or minus 1.96 standard errors across languages. They are descriptive and are not adjusted for multiple testing.

The demographic patterns vary across regions. The clearest descriptive pattern is the positive age by sex contrast in four regions, but these results do not establish a general demographic difference in colour perception.

Do the predictors improve held-out speaker predictions enough to justify complexity?



Question 5: what is the prediction target?

Target

For a speaker from a language already represented in the survey, predict that speaker's continuous near-split or compactness score from information available without using that speaker's own naming responses.

- The held-out unit is one speaker. The evaluation uses leave-one-speaker-out prediction.
- Age and recorded sex are individual-level predictors.
- Recurrent richness is calculated from the other speakers in the same language, $R_{\ell(s),-s}$.
- The speaker's own term count is excluded from a pre-response model because it is only known after the naming task.

PRESS: leave-one-out prediction

Let n be the number of observations, y_i the observed response for row i, and $\bar{y} = n^{-1}\sum_{i=1}^{n} y_i$ their sample mean. Let $\hat{y}_{i,(-i)}$ be the prediction for row i from the same OLS formula refitted after omitting that row. The leave-one-out error is $e_{i,(-i)} = y_i - \hat{y}_{i,(-i)}$.

$$PRESS = \sum_{i=1}^{n} e_{i,(-i)}^2 = \sum_{i=1}^{n} (y_i - \hat{y}_{i,(-i)})^2.$$

Two scale-aware summaries are

$$R^2_{PRESS} = 1 - \frac{PRESS}{\sum_{i=1}^{n}(y_i - \bar{y})^2}, \qquad RMSE_{LOO} = \sqrt{\frac{PRESS}{n}}.$$

For a model comparison, let $C_{\text{candidate}}$ and C_{simpler} be values of the same criterion for the candidate and simpler model, fitted to the same response and rows. Define

$$\Delta C = C_{\text{candidate}} - C_{\text{simpler}}.$$

Negative ΔAIC, ΔBIC, or $\Delta PRESS$ favours the candidate on that criterion.

Lower PRESS/RMSE$_{LOO}$ is better and negative R^2_{PRESS} means held-out predictions are worse than the mean benchmark. The ridge/lasso language-mean models below use repeated nested cross-validation instead.

Individual-score prediction: model ladder

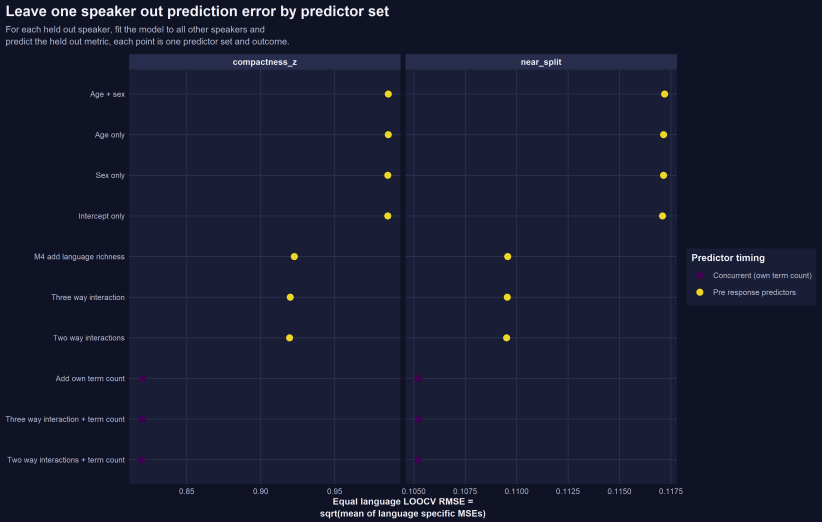
Let Y_s be either the speaker's near-split score or standardised compactness. Let A_s be age in decades, $S_s = 1$ for recorded male and 0 for recorded female, and $R_{\ell(s),-s}$ be recurrent richness computed without speaker s.

Pre-response prediction sequence

$$
\begin{aligned}
P_0: &\quad Y_s = \beta_0 + \beta_A A_s + \beta_S S_s + \varepsilon_s, \\
P_1: &\quad P_0 + \beta_R R_{\ell(s),-s}, \\
P_2: &\quad P_1 + \beta_{AR} A_s R_{\ell(s),-s} + \beta_{SR} S_s R_{\ell(s),-s} + \beta_{AS} A_s S_s, \\
P_3: &\quad P_2 + \beta_{ASR} A_s S_s R_{\ell(s),-s}.
\end{aligned}
$$

Concurrent comparison, not pre-response prediction

A separate sequence adds the speaker's own term count K_s to P_1, then considers the same interaction blocks. This can describe the completed naming partition, but K_s is unavailable before the task.



Leave-one-speaker-out prediction for near-split and compactness. The figure reports equal-language RMSE, averaging language-specific squared errors so each language has equal weight. Yellow points use pre-response predictors; purple points additionally use the held-out speaker's own term count, so they are concurrent rather than prospective.

Results: held-out speaker prediction

Pre-response model	Near-split pooled RMSE	R^2_{PRESS}	Compactness pooled RMSE	R^2_{PRESS}
Age + sex (P_0)	0.1191	−0.0025	1.0013	−0.0022
Add peer richness (P_1)	0.1115	0.1211	0.9229	0.1486
Add two-way terms (P_2)	0.1114	0.1224	0.9200	0.1540
Add three-way term (P_3)	0.1114	0.1219	0.9204	0.1532

- Richness provides the main pre-response predictive gain and the interaction terms add very little.
- Adding leave-one-speaker-out richness reduces RMSE by about 6.4% for near-split and 7.8% for compactness relative to age + sex.
- Two-way interactions add only about 0.08% and 0.32% further RMSE improvement, respectively. The three-way model is slightly worse.

The table reports pooled-speaker RMSE. The adjacent figure reports equal-language RMSE, averaging language-specific squared errors so each language has equal weight.

Concurrent comparison: adding the speaker's own term count

Why this is a separate comparison

The speaker's own term count K_s is calculated from the same naming responses as the outcomes. It is available after the task, so this is a concurrent explanatory comparison, not a pre-response forecast.

Concurrent model	Near-split RMSE	R^2_{PRESS}	Compactness RMSE	R^2_{PRESS}
Add own term count (C_1)	0.1070	0.1904	0.8304	0.3107
Add two-way terms (C_2)	0.1069	0.1912	0.8294	0.3125
Add three-way term (C_3)	0.1070	0.1908	0.8297	0.3119

The own-term-count addition substantially improves the descriptive fit, particularly for compactness. The interaction additions change prediction only slightly. These scores must not be presented as prospective prediction results.

Regression diagnostics: what we checked and why

The following residual, Q-Q and influence plots are from the individual-level OLS fits and the available language-level OLS reference fits. They do not diagnose the ridge/lasso models used for the primary language-mean prediction comparison.

Residuals vs fitted

We looked for curvature, funnels and systematic patterns. Curvature can indicate a mean formula that misses structure, while a funnel can suggest non-constant variance.

Leverage and Cook's distance

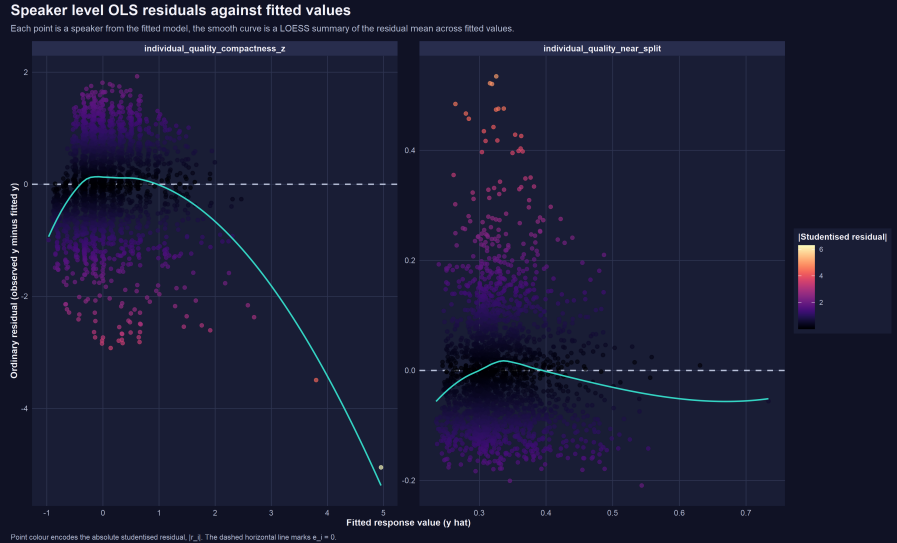
We flagged observations with unusual predictor values, large studentised residuals or high Cook's distance.

Normal Q-Q

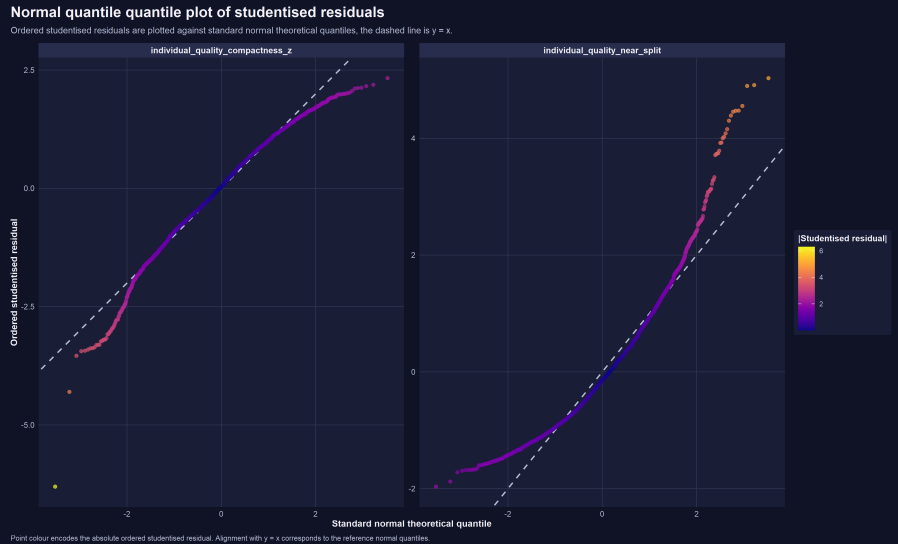
We checked how closely ordered studentised residuals followed the normal reference line. Departures in the tails matter for normal-theory inference, especially in small samples.

Collinearity and VIF

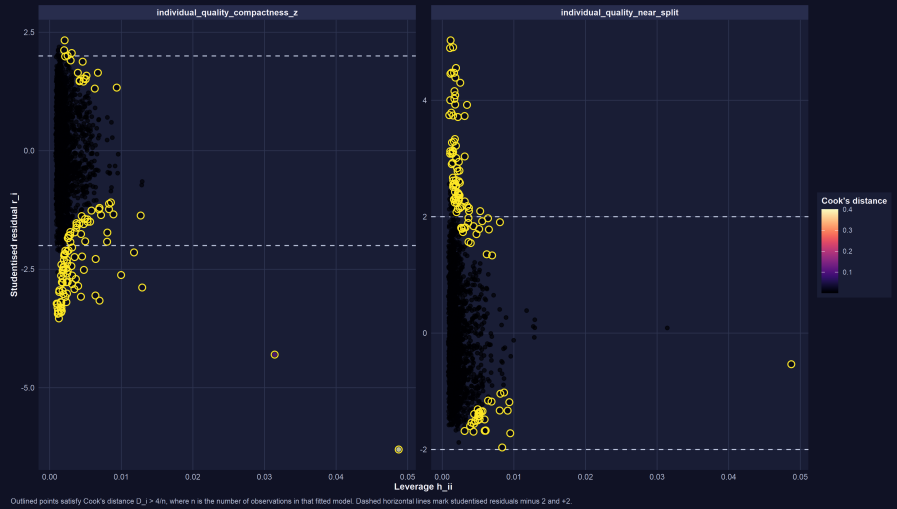
For the simple primary vocabulary sets, VIFs are 3.64 and 3.61 at language level and 3.82 and 3.80 for speaker peer predictors. The interaction-expanded designs are different: VIF reaches 49.03 in the individual near-split model and 9.92 in the language demographic interaction model.



Residuals are plotted against fitted values with a smooth trend. The clearest departure is strong curvature for individual-level compactness, including large negative residuals at high fitted values. Individual near-split also curves mildly. A funnel would suggest changing variance; curvature instead signals a possible mean-specification problem.



Ordered studentised residuals are compared with normal quantiles. Individual near-split has a pronounced upper-tail departure, while individual compactness has a marked lower-tail outlier. Normal-theory inference therefore needs caution.



Outlined points exceed the Cook's-distance screening rule and dashed lines mark studentised residuals of -2 and $+2$. Several individual-level cases are flagged and some language-level points have high leverage. A flag is not a deletion rule; no influence-deletion sensitivity result is reported.

Main conclusions: what the regressions support

1. **Mean partition geometry is outcome-specific.** In language-level nested CV, adding mean individual term count slightly worsens near-split prediction but reduces compactness RMSE by about 0.159 to 0.167. The M_0 scores are unavailable, so this comparison is M_2 versus M_1 only.

2. **Tail associations depend on adjustment.** The richness coefficients change markedly after adding mean individual term count. The lower near-split association becomes close to zero, while both compactness-tail coefficients reverse sign.

3. **Demographic complexity is hard to justify.** Some language-specific age/sex terms improve nested in-sample fit, but AIC, BIC and PRESS penalise the large expanded models. Two-way richness interactions give small gains; three-way terms do not.

4. **Regional richness patterns are concentrated.** The largest conditional slopes are green-to-cyan (0.0118) and blue-to-violet (0.0083). Both pass the simple Bonferroni threshold across eight regions, but the result remains exploratory.

5. **Richness helps individual prediction.** Leave-one-speaker-out richness reduces continuous-score RMSE by about 6.4% for near-split and 7.8% for compactness. Interactions add very little further improvement.

Limitations and scope of the conclusions

Measurement and sampling

- ΔE_{76} is a geometric proxy, not an individual perceptual threshold. The 330 chips do not cover the full visible-colour space.
- The 300-response eligibility threshold and the recurrent-term threshold are operational choices. Alternative thresholds could be checked.
- Age and recorded-sex metadata are incomplete and observational.

Modelling and inference

- Speakers from the same language may have correlated errors. No additional cluster-robust covariance estimator is used.
- OLS residual plots show curvature and tail departures. Homoscedasticity is not established by the reported diagnostics.
- Recurrent richness and mean term count share information; the compactness coefficient conditional on term count is not a marginal or causal effect.

Additional views of subgroup gaps and prediction



Supplement: language-level subgroup gaps

These contrasts summarise age and recorded-sex differences within each language and relate the gap to recurrent richness. They are descriptive language-level summaries, not the same estimand as the speaker-level interactions in Question 3.

Metric 2c: alignment with focal colours

For speaker s, term t, and chip i, use the category set I_{st} and centroid $\boldsymbol{\mu}_{st}$ defined earlier. Let $\mathcal{F}_s$ be the set of usable pairs (t, i) in which speaker s selected chip i as a focal example of term t and $|I_{st}| > 0$.

Mean focal-to-centroid distance

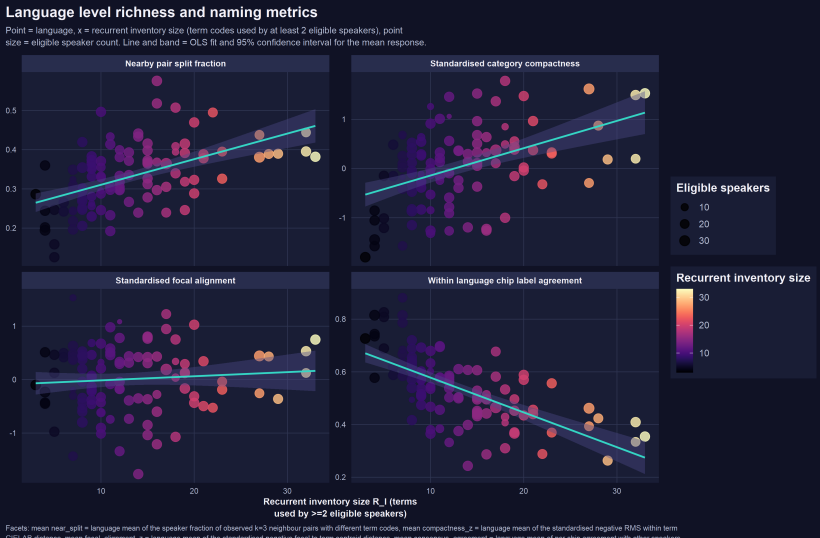
For $|\mathcal{F}_s| > 0$, define

$$D_s^{\text{focal}} = \frac{1}{|\mathcal{F}_s|} \sum_{(t,i)\in\mathcal{F}_s} \|\mathbf{x}_i - \boldsymbol{\mu}_{st}\|_2.$$

Let $V_s = -D_s^{\text{focal}}$, and let $\bar{V}, s_V$ be its mean and sample standard deviation over eligible speakers with defined scores. Then

$$Z_s^{\text{focal}} = \frac{V_s - \bar{V}}{s_V}.$$

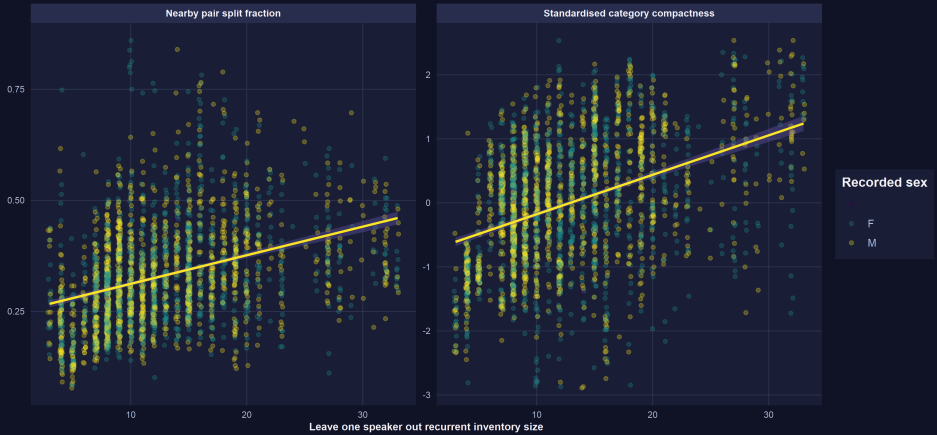
A high score means a speaker's selected focal chips lie closer to the centroid of chips they name with the corresponding term. This measures alignment between two WCS tasks.



Each point is a language, sized by eligible-speaker count. This is a descriptive overview of near-split, compactness, focal alignment and cross-speaker agreement. Only near-split and compactness are formal Question 1 outcomes.

Speaker level partition metrics against language derived vocabulary

Each point is one speaker, colour = recorded sex. The line is the OLS fit of the metric on the plotted richness variable, with a 95% confidence band for the mean response. Horizontal jitter only separates overlapping points.



Facets: near_split = fraction of observed k=3 nearest neighbour pairs receiving different term codes, compactness_z = standardised negative RMS distance from named chips to their own term centroid.

Descriptive speaker-level relationship between individual term count and partition metrics. This provides context for why term count is a concurrent descriptor rather than a pre-response predictor.

How the subgroup-gap analysis was constructed

Age-group contrast

Let age_s be recorded age in years. The cut-off $m_{\text{age}} = 34$ is the median age in the analysis sample: $\text{age}_s \leq m_{\text{age}}$ defines the younger group and $\text{age}_s > m_{\text{age}}$ the older group. For language ℓ, let $\bar{Q}_{\ell,O}$ and $\bar{Q}_{\ell,Y}$ be the selected outcome mean (or tail proportion) in the older and younger groups. Define

$$G_{\ell,\text{age}} = \bar{Q}_{\ell,O} - \bar{Q}_{\ell,Y}.$$

Recorded-sex and joint contrasts

Let $\bar{Q}_{\ell,g,h}$ denote the selected outcome mean (or tail proportion) for age group $g \in \{O, Y\}$ and recorded sex $h \in \{M, F\}$. Let $\bar{Q}_{\ell,M}$ and $\bar{Q}_{\ell,F}$ be the corresponding overall sex-specific means/proportions, pooling age groups. Define

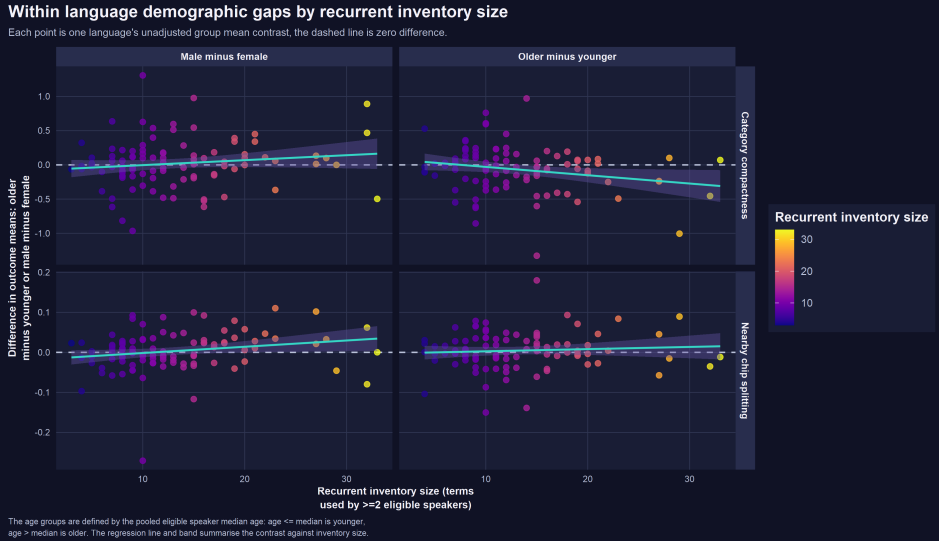
$$G_{\ell,\text{sex}} = \bar{Q}_{\ell,M} - \bar{Q}_{\ell,F},$$

$$G_{\ell,AS} = (\bar{Q}_{\ell,O,M} - \bar{Q}_{\ell,O,F}) - (\bar{Q}_{\ell,Y,M} - \bar{Q}_{\ell,Y,F}).$$

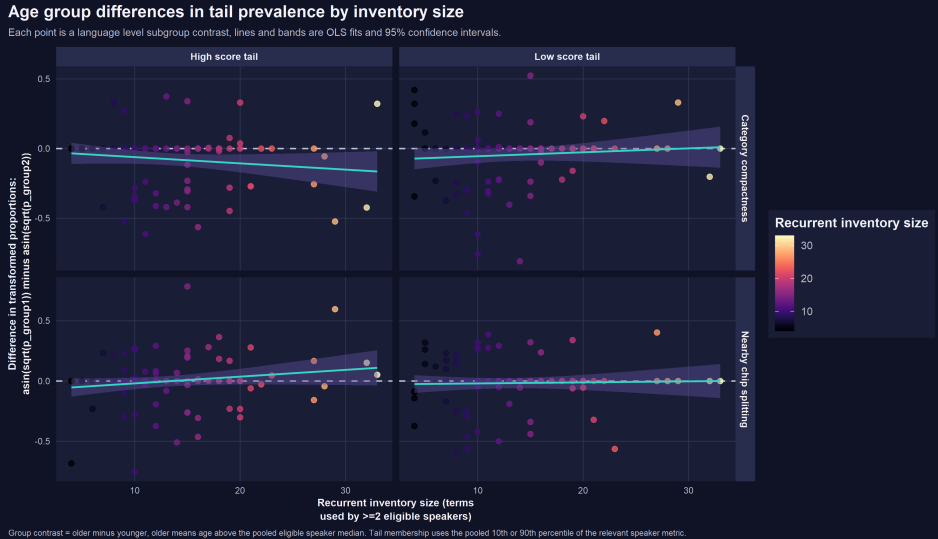
Language-level gap-regression ladder (fit separately for each gap):

- C_0: gap $\sim \log(n_\ell)$ (sample-size baseline).
- C_1: add recurrent inventory richness R_ℓ.
- C_2: add mean individual term count $\bar{K}_\ell$.

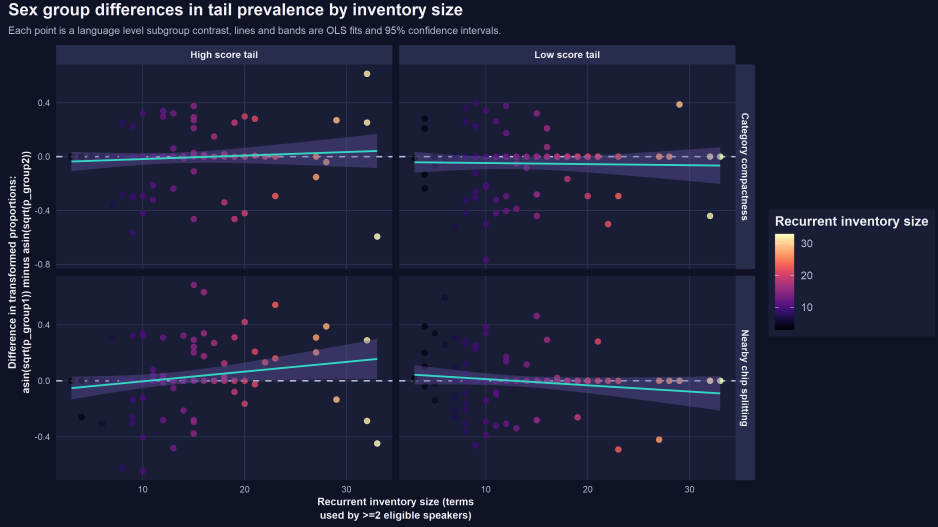
For low/high-tail gaps, the outcome is the difference in the transformed tail proportions. The median split is used only for these language-level subgroup summaries. The earlier age-slope models use age continuously in decades. These aggregate gap regressions are not the same as the speaker-level interaction models above.



Each point is a language-level age, recorded-sex or joint subgroup contrast plotted against recurrent richness. The vertical axis is a group difference, not an individual score. These trends are descriptive.



Each point is the language-level difference in lower- or upper-tail prevalence between older and younger speakers. Panels separate the metric and tail. These are descriptive subgroup contrasts.



Each point is the language-level difference in tail prevalence between recorded-sex groups. Direction and uncertainty vary across metrics and tails.

What do the subgroup-gap plots show?

- **Continuous outcomes.** The fitted language-level gaps are shallow. For compactness, the male-minus-female line rises slightly with richness while the older-minus-younger line slopes down. For near-split, both fitted lines are close to flat or only slightly positive.
- **Tail outcomes.** Age and sex gap trends vary by metric and tail. Some fitted directions differ between the lower and upper compactness tails, while several near-split gaps stay close to zero. The language-level points are widely scattered and the confidence bands are broad.
- **What we can conclude.** These plots suggest no single demographic gap that changes consistently with richness across all outcomes. They are descriptive regressions of language-level contrasts. The C_0 to C_2 gap-model sequence is defined above, but this deck does not report its AIC, BIC or PRESS comparisons, so we do not declare a formal winner for those models.

Supplement: predicting lower- and upper-tail membership

Target. For each speaker, define a binary indicator for membership in the lower or upper decile of near-split or compactness. This is a classification target, distinct from the language-level tail proportions in Question 2.

Binary GLM

For each metric and tail, fit

$$\Pr(T_s = 1 \mid \mathbf{x}_s) = \frac{\exp(\mathbf{x}_s^\top \beta)}{1 + \exp(\mathbf{x}_s^\top \beta)}.$$

The predictor ladder starts with an intercept, then age, recorded sex, age plus sex, leave-one-speaker-out richness, two-way interactions and the three-way interaction.

- Evaluate by leave-one-speaker-out prediction.
- Recalculate the 10th/90th percentile threshold within each training fold to avoid using the held-out speaker to define the target threshold.
- Compare held-out negative log-likelihood (NLL) and classification error. NLL is informative here because the tail class is only about 10% of speakers.

Supplement: best held-out tail-membership models

Metric	Tail	Best model	Equal-language held-out NLL
Near-split	Low	Age + sex + peer richness (T_4)	0.2846
Near-split	High	T_4 + two-way interactions (T_5)	0.2997
Compactness	Low	Age + sex + peer richness (T_4)	0.2989
Compactness	High	T_4 + two-way interactions (T_5)	0.2894

All models predict 2,355 held-out speaker outcomes. Classification error is close to 10%, which is expected for a tail indicator with roughly 10% prevalence. The held-out NLL is therefore the more informative comparison of predicted probabilities.

Supplement: predicting the exact term on a held-out chip

Different target. This analysis predicts the exact term code assigned to a chip, rather than a speaker's continuous partition score.

- Hold out a reproducible 20% set of chip IDs and retain the same held-out chip set for every model.
- Fit language-specific one-versus-rest logistic models for each term: that term versus all other term codes.
- Use CIELAB coordinates as the baseline predictors. The ladder then adds age, recorded sex, training-chip vocabulary size, and an age-by-sex interaction.
- Predict the term with the largest fitted one-versus-rest probability and evaluate exact agreement on the common held-out responses.

The holdout is by chip ID within speakers already observed. It tests spatial interpolation for these speakers.

Supplement: exact-label accuracy on held-out chips

Model	Added information	Exact accuracy
L_0	CIELAB coordinates only	61.09%
L_1	Add age	61.66%
L_2	Add recorded sex	62.11%
L_3	Add training-chip vocabulary size	62.74%
L_4	Add age-by-sex interaction	63.06%

- 154,921 held-out speaker-chip responses from 2,355 speakers and 108 languages are evaluated.
- The fraction of true labels with a model supported by the training chips is 98.17%.
- The full ladder raises exact accuracy from 61.09% to 63.06%, a gain of 1.97 percentage points over the CIELAB-only baseline.

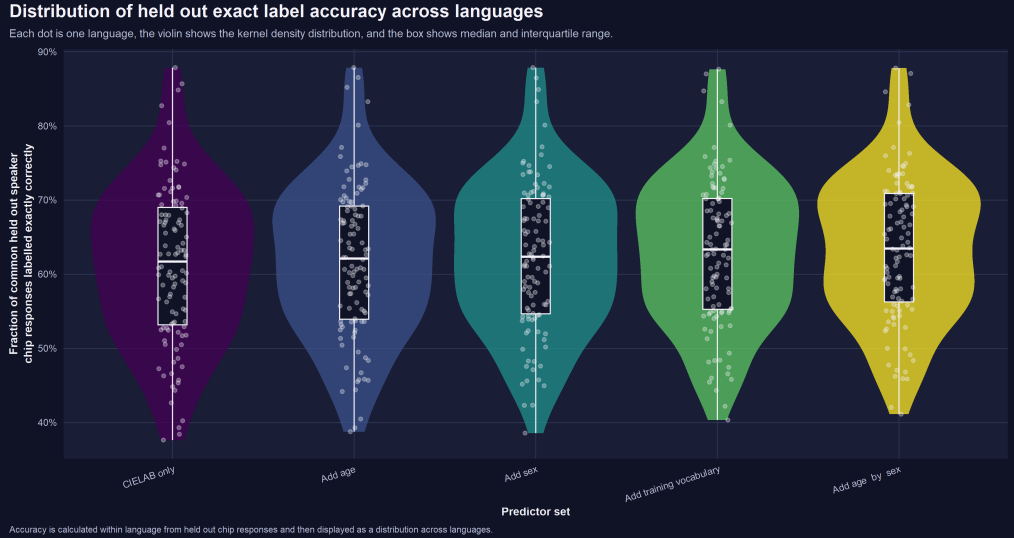
Exact term code accuracy on held out chip responses

Each point is a model, accuracy is the fraction of common held out responses for which the predicted term code equals the observed code.

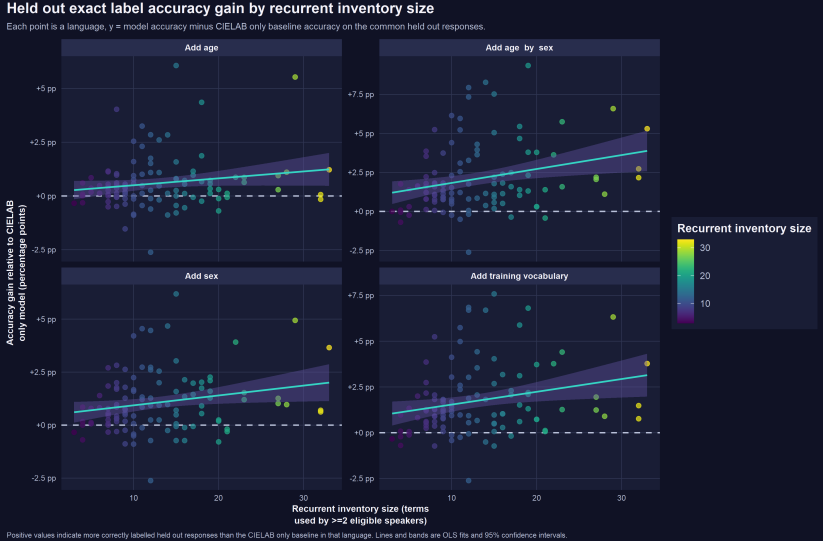


The same set of held out speaker chip responses is used for every model in this comparison. The predicted label is the term class with the largest fitted one versus rest probability.

Exact term-code accuracy on the common held-out chip responses rises from 61.09% with CIELAB coordinates alone to 63.06% with the full predictor set.



Each dot is one language. The distributions show language-to-language variation in exact held-out-chip accuracy under each predictor set. A pooled average conceals substantial heterogeneity across languages.



Each point is a language. The vertical axis is exact-label accuracy gain relative to the CIELAB-only model. Lines are descriptive OLS trends with 95% confidence intervals. Gains vary across languages; recurrent richness is then tested in a separate language-level regression.

Supplement: does richness explain differences in exact-label gain?

For each language, the response is its change in held-out exact-label accuracy relative to the CIELAB-only baseline. This is a separate language-level regression because recurrent richness is constant within a language-specific label model.

Accuracy gain	Model step	ΔAIC	ΔBIC	ΔPRESS
Add age	Add richness	−1.36	+1.32	−0.00009
Add sex	Add richness	−3.36	−0.67	−0.00061
Add training vocabulary	Add richness	−5.27	−2.58	−0.00186
Add age-by-sex	Add richness	−7.45	−4.77	−0.00336

The richness step improves AIC, BIC and PRESS for each gain outcome shown. For the training-vocabulary gain, its nested F-test has $p = 0.0080$. The corresponding PRESS-based predictive R^2 remains negative, so prediction of the language-to-language gain itself is still weak.



For each added predictor step, bars show the fraction of term-specific one-versus-rest GLMs whose AIC, BIC or both decreased. These criteria are summarised across models; they are not summed across overlapping one-versus-rest outcomes.

Supplement: prediction for a genuinely unseen language

Here the held-out unit is an entire language. The target is its mean near-split or mean compactness score. Unlike Question 1, recurrent richness is deliberately excluded because a new language's inventory is not known before collecting its naming responses.

Leave-one-language-out model ladder

$$N_0 : \text{intercept only},$$
$$N_1 : \log(n_\ell),$$
$$N_2 : \log(n_\ell) + \text{mean age summary},$$
$$N_3 : \log(n_\ell) + \text{mean age summary} + \text{sex composition}.$$

Each language is held out once. PRESS compares the resulting out-of-language predictions.

This is not the same as Question 1's held-out-language ridge/lasso analysis, which uses the held-out language's observed richness and mean term count as predictors.

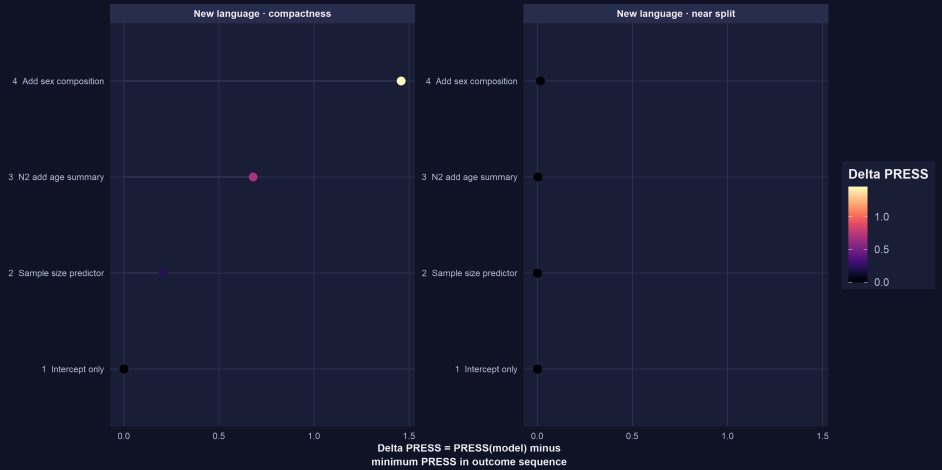
Supplement: unseen-language prediction is weak

Outcome	Model	PRESS	R^2_{PRESS}	Reading
Near-split	N_0 intercept	0.7073	−0.0188	Baseline
Near-split	N_1 + sample size	0.7071	−0.0185	Tiny PRESS gain
Near-split	N_2 + age summary	0.7108	−0.0238	Worse
Near-split	N_3 + sex composition	0.7235	−0.0421	Worse
Compactness	N_0 intercept	66.6594	−0.0188	Best
Compactness	N_1 + sample size	66.8678	−0.0220	Worse
Compactness	N_2 + age summary	67.3394	−0.0292	Worse
Compactness	N_3 + sex composition	68.1178	−0.0411	Worse

The PRESS-based predictive R^2 is negative for every model. Sample size gives only a negligible near-split improvement, while age and sex summaries do not improve prediction. None of the added blocks is preferred by AIC or BIC.

Leave one language out prediction error

Each language is omitted once, the model fitted to the remaining languages predicts that language's mean outcome.



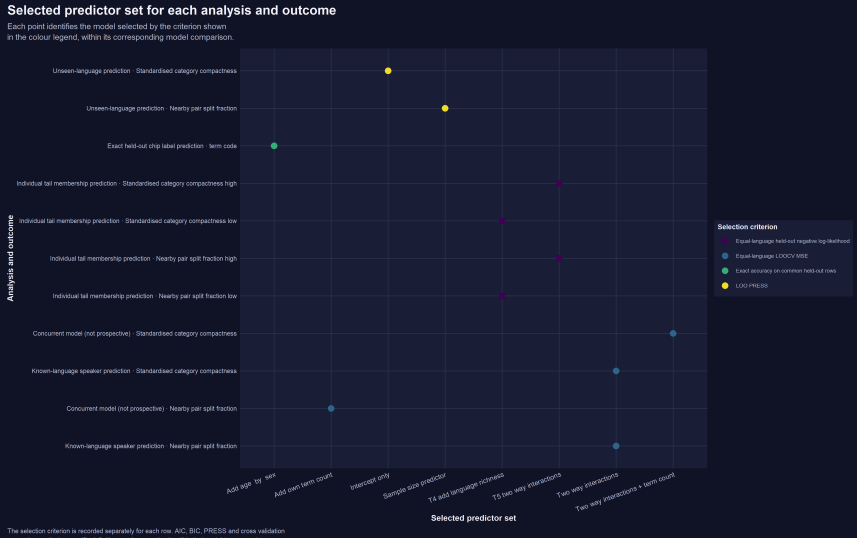
PRESS = sum over held out languages of (observed language mean minus predicted language mean)^2. Predictors are log eligible speaker count, mean age and female proportion, which can be summarised without observing the held out language's colour naming.

Leave-one-language-out prediction using only sample-size and demographic summaries available without the new language's naming inventory. The predictive R^2 values are negative, indicating no useful improvement over the mean benchmark in these models.

Supplement: choosing the best predictor by target

The prediction targets are not interchangeable. The preferred model depends on whether the target is a continuous speaker score, tail membership, an exact chip label, or a mean score for a genuinely unseen language.

- For continuous scores in an already represented language, leave-one-speaker-out richness gives the main gain. Two-way interactions add little.
- For lower/upper-tail membership, peer richness improves held-out probability scores; the best interaction order differs by tail.
- For exact held-out chip labels, the full age-by-sex model gives the highest common-row accuracy, but this is within-speaker spatial interpolation.
- For a genuinely unseen language, richness is unavailable in advance and sample-size/demographic summaries give weak PRESS performance.



This summary maps the best-performing predictor set to each outcome and prediction target. Read each row with its criterion and holdout unit; do not compare raw scores across different targets.

References and data provenance

1. Berlin, B. & Kay, P. (1969). *Basic Color Terms: Their Universality and Evolution*. University of California Press.
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